

1. System Requirements Specification

1.1 The 2 Functional Requirements (FR)

- **FR-1: Route Schedule Browsing & Interactive Seat Selection** *Description:* The system shall allow commuters and dispatchers to browse intercity bus schedules (origin, destination, departure time, and base fare) and dynamically select an available seat through a visual 2x2 bus cabin layout.
- *Inputs:* Route selection from dropdown, travel date (YYYY-MM-DD), and visual seat grid click.
- *Expected Output:* Real-time seat state toggling, passenger qualification checks, and live fare calculation.

FR-2: Booking Processing, Ticket Issuance & Lifecycle Cancellation

- *Description:* The system shall process confirmed seat reservations into persistent SQLite records, generate tamper-resistant verification codes, and support ticket cancellation with immediate seat inventory release and audit logging.
- *Inputs:* Passenger details (Name, Contact, Classification, ID proof) and Ticket ID / Reference Code lookup.
- *Expected Output:* Ticket record created in **tickets** table; cancellation updates status to **Cancelled** and frees the seat for subsequent passengers.

2. The 2 Non-Functional Requirements (NFR)

- **NFR-1: Security, Validation & Tamper Resistance** *Quality Attribute: Security & Data Integrity*
- *Specification:* All passenger inputs must undergo strict sanitization. Contact numbers must conform to standard mobile telephone formats (09XXXXXXXX or +639XXXXXXXX). Concession fare discounts require mandatory verification of official institutional/government ID credentials (minimum 4 characters). Every ticket is issued with an 8-character cryptographically salted SHA-256 reference code (BTK-[HEX]) to prevent ticket counterfeiting.

NFR-2: Maintainability, Clean Architecture & Double-Booking Prevention

- *Quality Attribute: Maintainability, Reliability & ACID Concurrency*
- *Specification:* The codebase strictly enforces Object-Oriented encapsulation (private _ attributes, @property getters, and @setter validation raising **TypeError** or **ValueError**) conforming to PEP 8 standards. At the database layer, an ACID-compliant SQLite schema with a partial unique index (**ON tickets (route_id, travel_date, seat_number) WHERE status = 'Confirmed'**) mathematically guarantees that no seat can ever be double-booked.

3. Unique System Use Case

Interactive Visual Seat Layout Matrix with Senior/PWD Priority Allocation & Concession Discount Engine

- **Use Case ID: UC-SMARTBUS-01**
- **Actor:** Passenger / Transit Booking Officer
- **Preconditions:** SQLite database is initialized with active routes and assigned fleet buses.
- **Core Flow:** The user selects an intercity route (e.g., **RT-101: Manila (Cubao) -> Baguio City**) and travel date.
- The system queries SQLite and dynamically renders a **2x2 Cabin Seating Matrix** featuring a center aisle: **Row 1 (Seats 1, 2, 3, 4)** are designated **Priority Seats**, color-coded in vibrant **Amber**.
- **Rows 2 to 6** are designated **Regular Seats**, color-coded in **Green** (Available) or **Red** (Booked).

When an available seat is clicked, it highlights in **Blue** ([✓]).

If a non-priority commuter selects a Row 1 priority seat, the system triggers a policy verification prompt.

The commuter selects their passenger category:

- **Regular:** Standard base fare charged.

- **Senior Citizen / PWD / Student:** The system enables the Concession ID entry field and automatically deducts a statutory **20% fare discount** (e.g., PHP 580.00 base fare yields a PHP 116.00 discount, for a net payable amount of PHP 464.00).

Upon confirming booking, the system atomically commits the reservation to SQLite, marks the seat as occupied (**[X]**), and issues the verified ticket.

Postconditions: The seat is locked against duplicate bookings in SQLite; the passenger receives a printable boarding pass.

4. System Architecture & Relational Schema

4.1 Layered Architecture Pattern

The system follows a Model-View-Controller (MVC) structure decoupled via the Repository/Singleton pattern:

Code Listing

```
+-----+
| PRESENTATION LAYER (View) |
| - Tkinter GUI Subclass (app.py: BusTicketingApp) |
| - Multi-Tab ttk.Notebook (Tabs 1, 2, 3) & ttk.Treeview Tables |
| - Interactive Seat Canvas/Grid & Modal Boarding Pass Dialog |
+-----+
| Event Triggers & Form Data
v
+-----+
| BUSINESS LOGIC & FACTORY LAYER |
| - TicketFactory (models/factory.py: Fare calc, concession checks) |
| - BusFactory (models/factory.py: Fleet cabin configuration) |
| - Domain Models: Route, Bus, Passenger, Ticket, TicketStatus |
+-----+
| Verified Domain Objects
v
+-----+
| DATA ACCESS & PERSISTENCE LAYER |
| - BusTicketingDatabase Singleton (models/database.py) |
| - Parameterized SQL Queries & Context-Managed Connections |
| - Partial Unique Indexes for Double-Booking Prevention |
+-----+
| ACID Transactions
v
+-----+
| SQLITE DATABASE ENGINE |
| - Local Relational File: 'bus_ticketing.db' |
+-----+
```

4.2 SQLite Relational Database Schema

Code Block: SQL

```
-- 1. Routes Table
CREATE TABLE routes (
  route_id TEXT PRIMARY KEY,
  origin TEXT NOT NULL,
  destination TEXT NOT NULL,
  distance_km REAL NOT NULL,
  base_fare REAL NOT NULL,
  departure_time TEXT NOT NULL,
  bus_id TEXT NOT NULL
);

-- 2. Buses Table
CREATE TABLE buses (
  bus_id TEXT PRIMARY KEY,
  bus_number TEXT NOT NULL,
  bus_type TEXT NOT NULL,
  total_seats INTEGER NOT NULL,
  layout_rows INTEGER NOT NULL,
  layout_cols INTEGER NOT NULL,
  priority_seats_json TEXT NOT NULL
);

-- 3. Passengers Table
CREATE TABLE passengers (
  passenger_id TEXT PRIMARY KEY,
  name TEXT NOT NULL,
  contact_number TEXT NOT NULL,
  passenger_type TEXT NOT NULL,
  discount_id TEXT
);

-- 4. Tickets Table (with Foreign Keys & Concession Audit)
CREATE TABLE tickets (
  ticket_id TEXT PRIMARY KEY,
  booking_reference TEXT UNIQUE NOT NULL,
  route_id TEXT NOT NULL,
  bus_id TEXT NOT NULL,
  passenger_id TEXT NOT NULL,
  seat_number INTEGER NOT NULL,
  travel_date TEXT NOT NULL,
  base_fare REAL NOT NULL,
  discount_amount REAL NOT NULL,
  final_fare REAL NOT NULL,
  status TEXT NOT NULL,
  booking_time TEXT NOT NULL,
  cancellation_time TEXT,
  cancellation_reason TEXT,
  FOREIGN KEY (route_id) REFERENCES routes(route_id),
  FOREIGN KEY (bus_id) REFERENCES buses(bus_id),
  FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id)
);

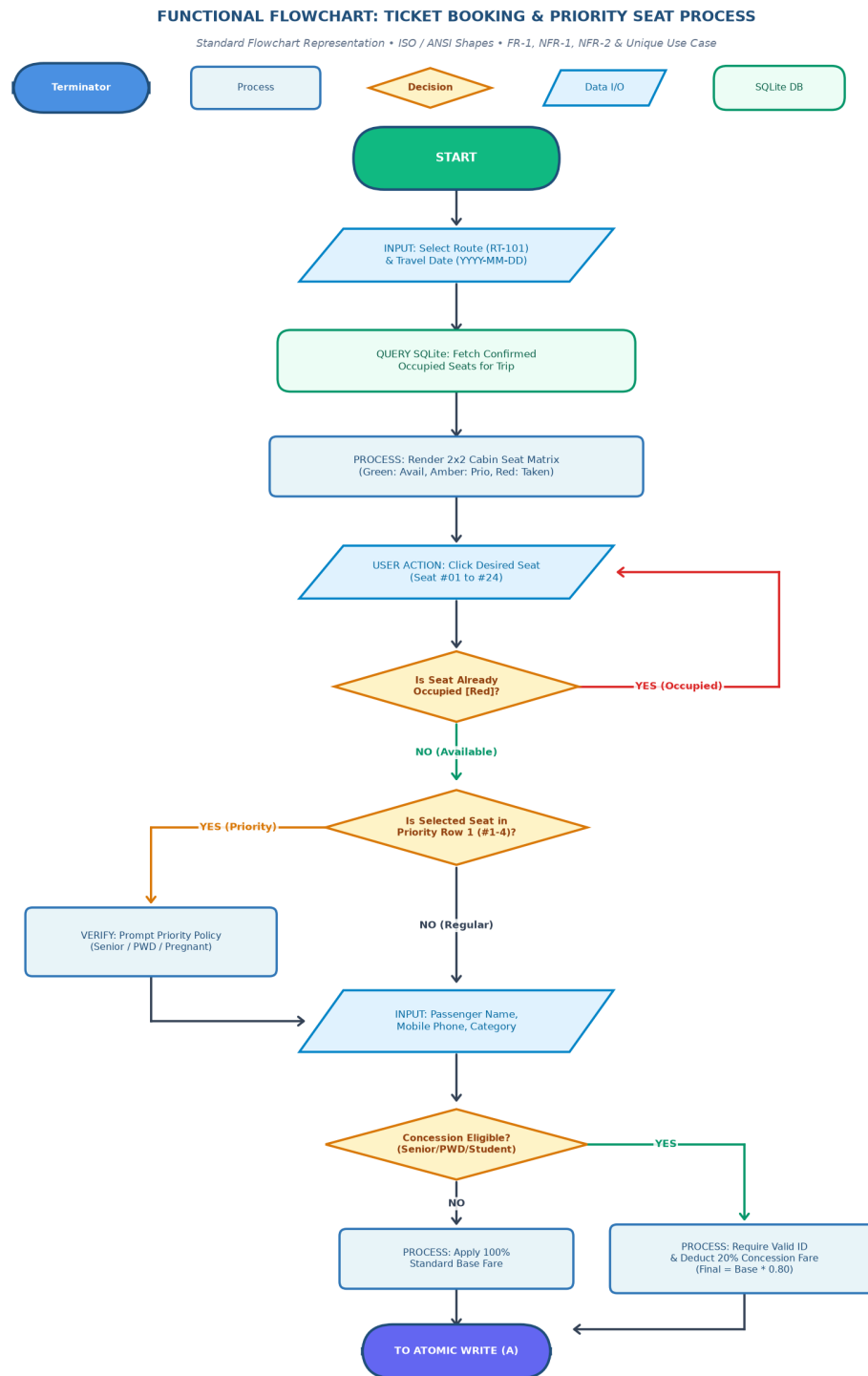
-- Strict Double-Booking Prevention (NFR-2)
CREATE UNIQUE INDEX idx_unique_active_seat
ON tickets (route_id, travel_date, seat_number)
WHERE status = 'Confirmed';
```

5. Process Flowcharts & Workflows

5.1 Standard Flowchart: Ticket Booking & Concession Allocation (FR-1 & Unique Use Case)

Standard flowchart utilizing ISO/ANSI flowchart shapes: **Terminator capsules** (Start/End), **Input/Output parallelograms**, **Process rectangles**, **Decision diamonds** (Yes/No branches), and **Database cylinders** (SQLite queries):

Figure: Generic Flowchart - Booking Process



Code Block: MERMAID

```

flowchart TD
    Start([User Initiates Booking]) --> SelectRoute[/Input: Select Route & Travel Date/]
    SelectRoute --> QueryDB[(Query SQLite for Booked Seats)]
    QueryDB --> RenderGrid[Render 2x2 Interactive Seat Grid  
Green: Available | Red: Booked | Amb...]
    RenderGrid --> ClickSeat[/User Clicks Desired Seat/]

    ClickSeat --> IsOccupied{Is Seat Already  
nOccupied?}
    IsOccupied -- Yes --> AlertOccupied[Show Warning: Seat Already Booked]
    AlertOccupied --> ClickSeat

    IsOccupied -- No --> CheckPriority{Is Seat in Priority Row?  
nRows 1 & 2}
    CheckPriority -- Yes --> PromptPriority[Verify Senior / PWD Eligibility]
    CheckPriority -- No --> SelectSeat[Highlight Seat Blue [✓]]
    PromptPriority --> SelectSeat

    SelectSeat --> EnterDetails[/Input: Name, Phone & Category/]
    EnterDetails --> SelectCategory{Select Classification}

    SelectCategory -- Regular --> CalcRegular[Base Fare Applied  
n0% Discount]
    SelectCategory -- Senior / PWD / Student --> RequireID[/Input: Concession ID/]
    RequireID --> CalcDiscount[Apply 20% Concession Deduction  
nFinal = Base * 0.80]

    CalcRegular --> SubmitBooking[Click Confirm & Issue Ticket]
    CalcDiscount --> SubmitBooking

    SubmitBooking --> ValidatePII{Validate PII Regex &  
nConcession ID Proof}
    ValidatePII -- Invalid --> DisplayError[Show Validation Error Alert]
    DisplayError --> EnterDetails

    ValidatePII -- Valid --> GenHash[Generate Salted SHA-256 Code  
nBTK-XXXXXX]
    GenHash --> AtomicInsert[(Begin SQLite Transaction  
nInsert Ticket & Passenger)]

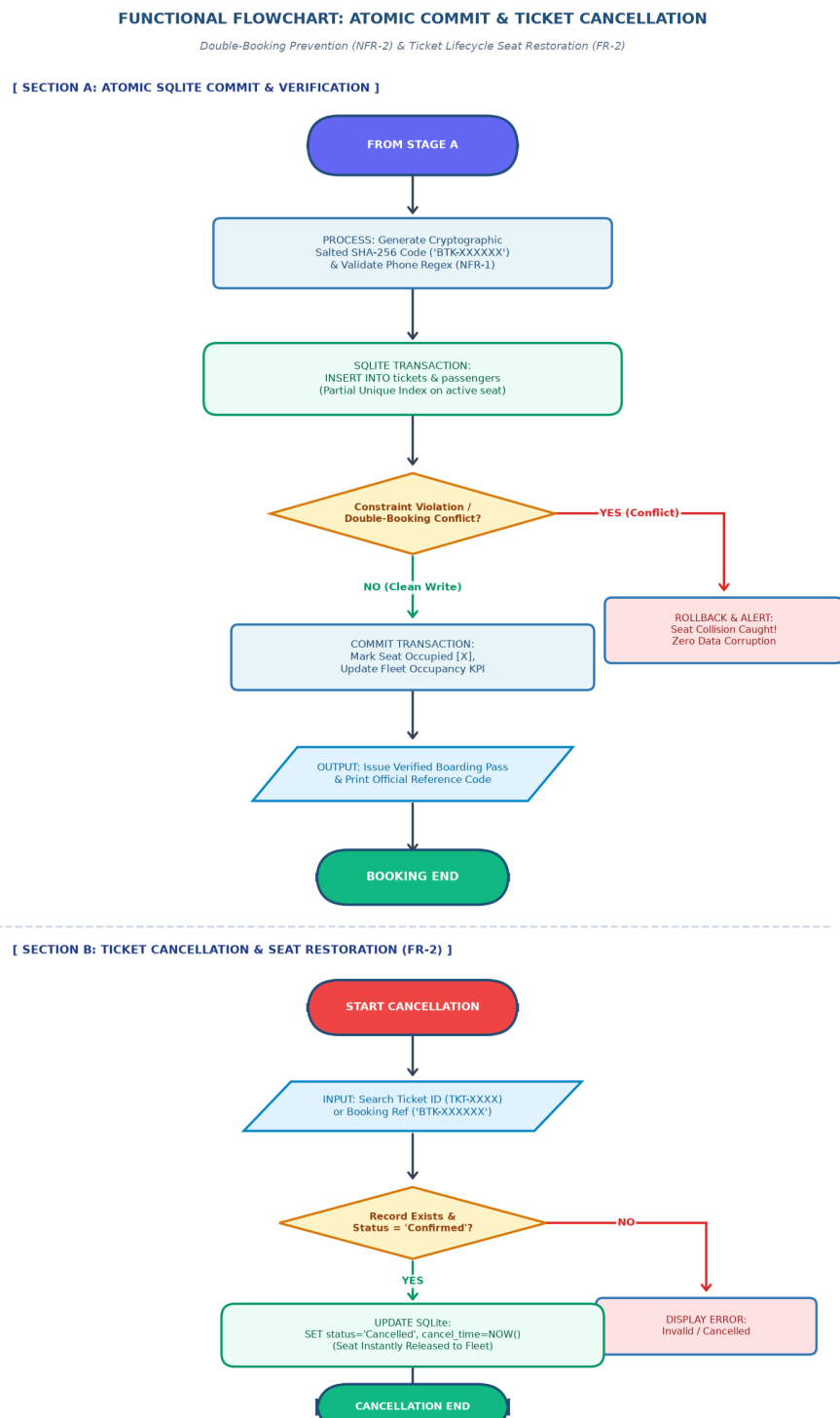
    AtomicInsert -- Collision Detected --> Rollback[Rollback Transaction & Alert Conflict]
    AtomicInsert -- Success --> CommitDB[Commit SQLite Transaction]
    CommitDB --> RefreshUI[Update Seat Grid to Red [X]  
nIncrement Analytics KPI & Issue Boarding ...]
    RefreshUI --> End([Booking Complete])

```

5.2 Standard Flowchart: Ticket Cancellation & Seat Release (FR-2)

Standard flowchart for lifecycle ticket cancellation and automatic seat restoration:

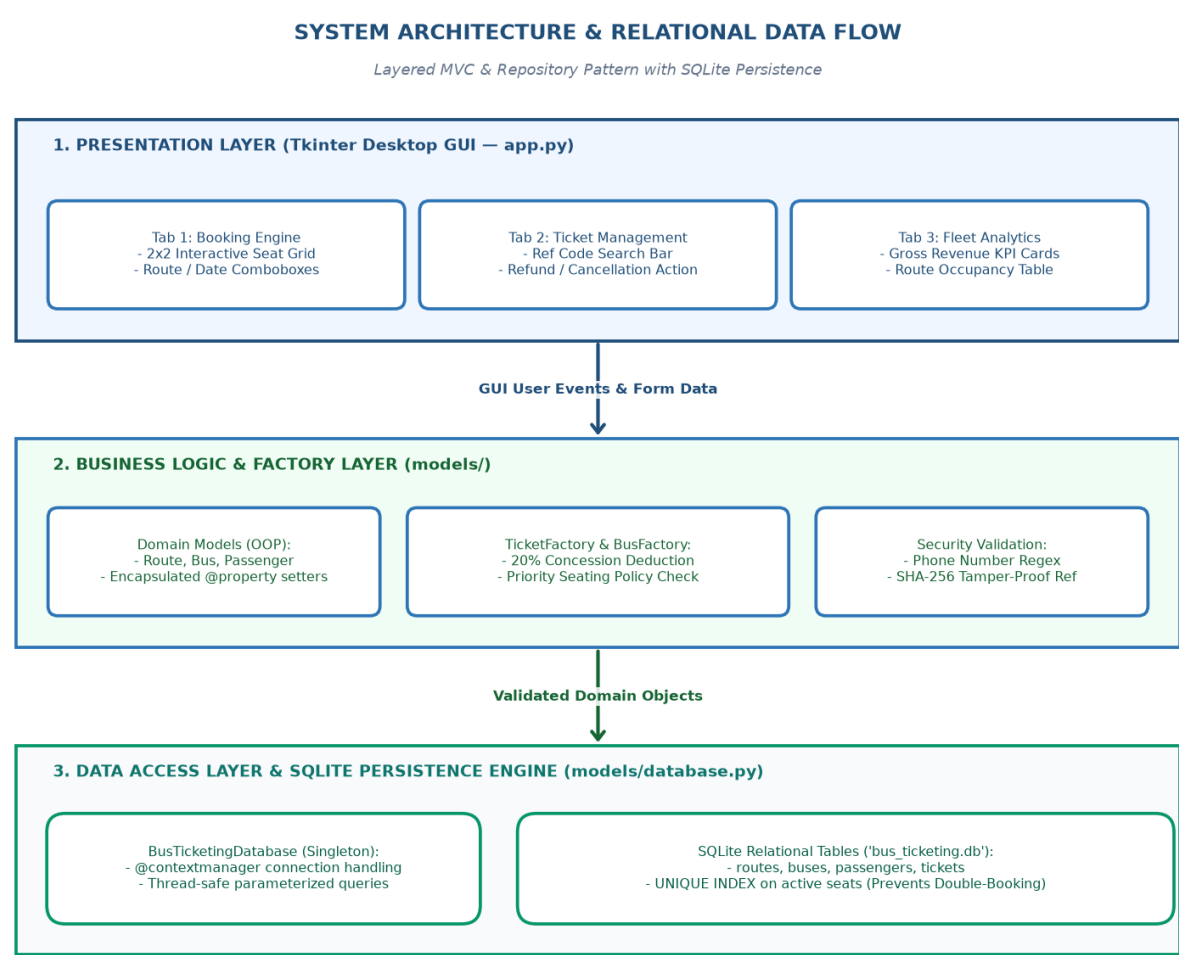
Figure: Generic Flowchart - Cancellation Process



5.3 High-Level Architecture & Relational Data Flow

Layered Model-View-Controller architecture and data flow through SQLite:

Figure: System Architecture & Data Flow



6. UI Wireframes & Visual Layouts

Tab 1 Wireframe: Book Tickets & Visual Seat Map

Figure: Tab 1 Wireframe

SmartBus Transit System — Desktop Booking Interface (Lab Practical)

1. Book Tickets & Visual Seat Map

2. Manage Bookings

3. System Analytics

1. Select Route & Trip

Route: [RT-101: Manila -> Baguio ▼]

Travel Date: [2026-09-28]

[Update Seat Grid]

=== ROUTE ITINERARY ===

Origin : Manila (Cubao)

Dest. : Baguio City

Distance : 245.0 km

Departs : 08:00 AM

Base Fare: PHP 580.00

=== FLEET BUS ===

Plate : DLTB-8012

Class : Standard AC

Capacity : 24 Seats

Priority : Row 1 (#1-4)

2. Interactive 2x2 Seat Matrix

AvailPrioritySelectedTaken

[DRIVER CAB / FRONT]

R1	01 [P]	02 [P]	03 [✓]	04 [P]
R2	05	06	07 [X]	08
R3	09	10	11	12 [X]
R4	13	14	15	16
R5	17	18	19 [X]	20
R6	21	22	23	24

3. Passenger & Concession

Full Name:
[Maria Santos]

Contact Number:
[09171234567]

Category:
[Senior Citizen (20% Off) ▼]

Concession / OSCA ID:
[OSCA-77491]

Base Fare : PHP 580.00

Discount : -PHP 116.00 (20%)

Total Due : PHP 464.00

[Confirm & Issue Ticket]

System Ready • Connected to SQLite 'bus_ticketing.db' • ACID Transactions Enabled

Tab 2 Wireframe: Manage Bookings & Audit Dossier

SmartBus Transit System • Laboratory Practical • SQLite Relational Engine

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Figure: Tab 2 Wireframe

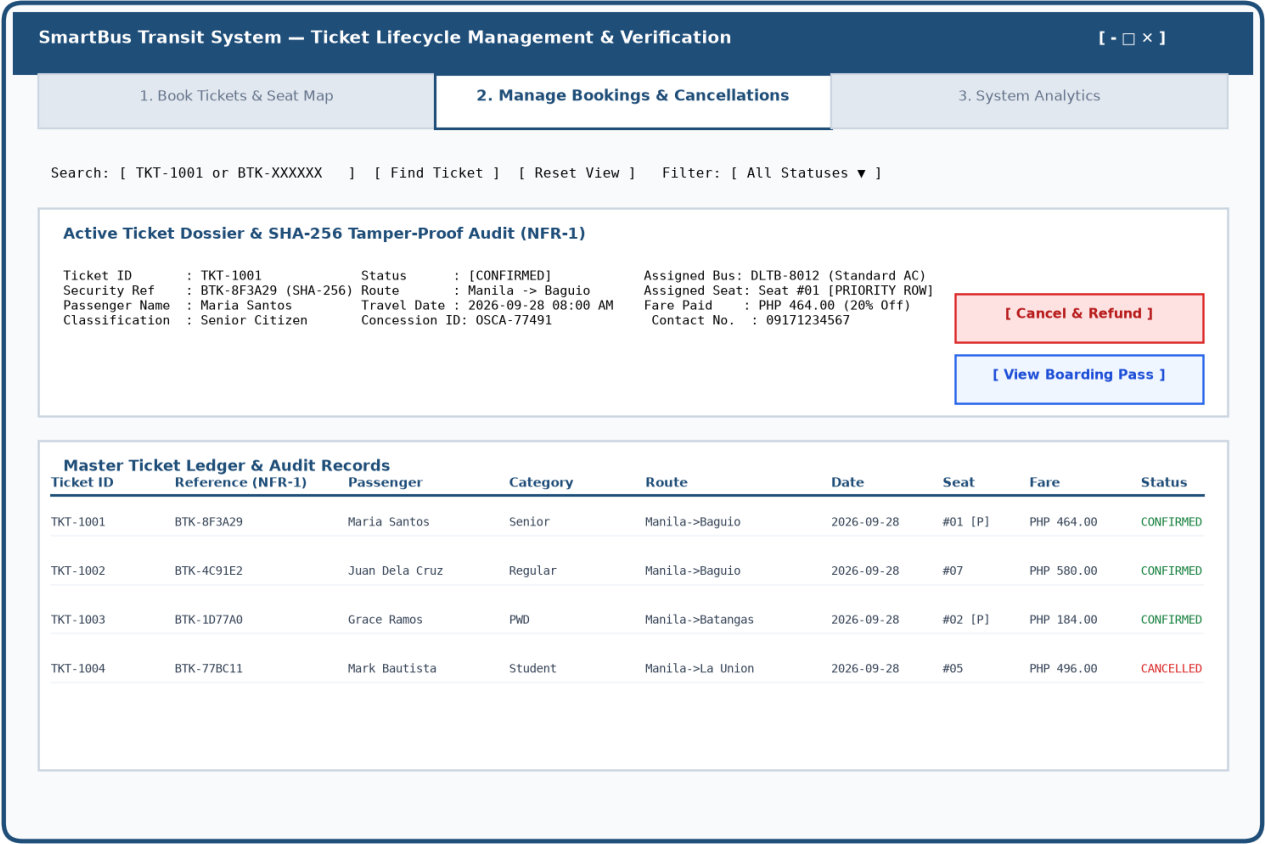
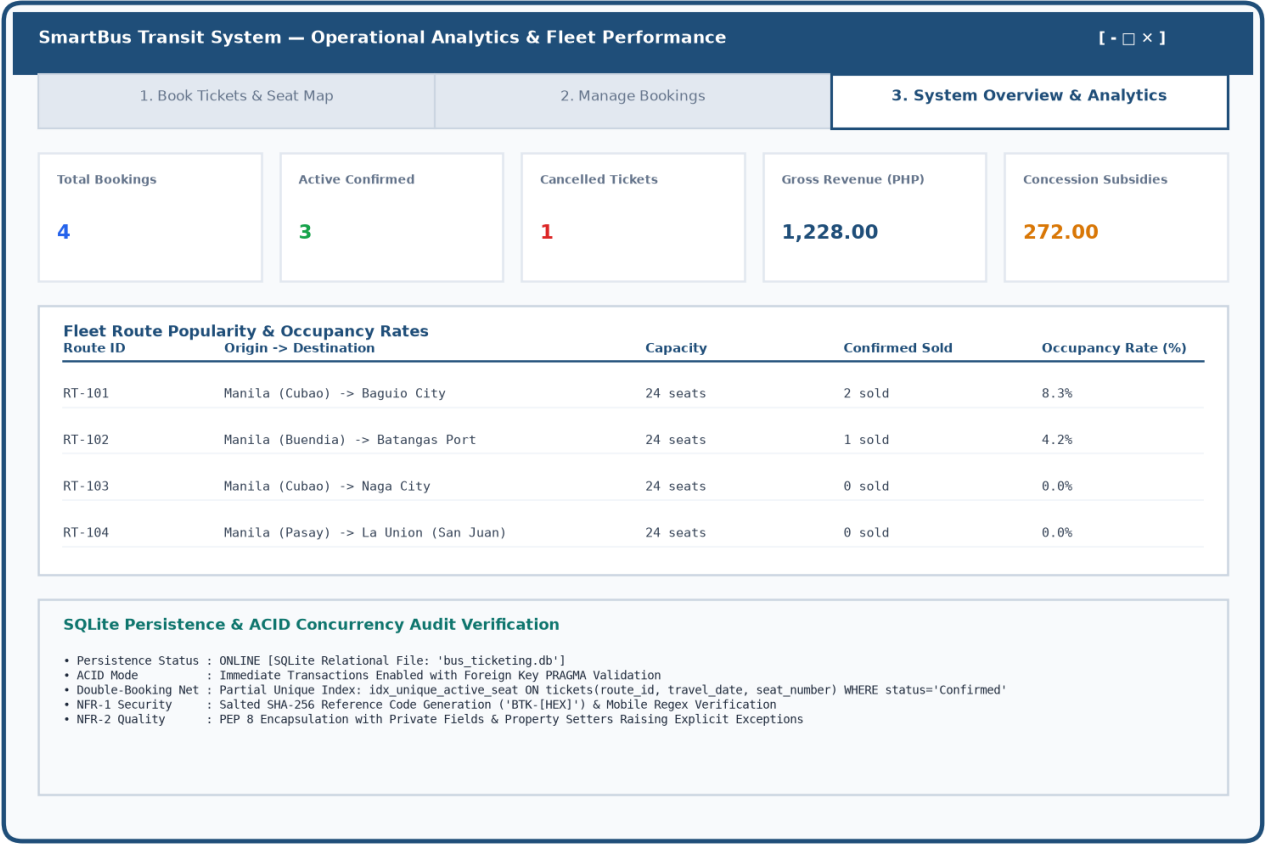


Figure: Tab 3 Wireframe



7. Live Interface Screenshots

Milestone & Scenario

01. Tab 1 Default View

Route selection, trip itinerary, and 2x2 visual seat map with Row 1 Priority seats.

SmartBus Transit System — Booking & Fleet Management

Tab: Dashboard | SQLite Backend | Concessions Priority Engine

1. Book Tickets & Visual Seat Map

2. Manage Bookings & Cancellations

3. System Overview & Fleet Analytics

4. Passenger & Concessions

Transit Route

RT-101 - Manila (Cubao) to Baguio City (08:00 AM)

Travel Date (YYYY-MM-DD): 2025-09-26

Update Seat Availability

Route & Bus Details

===== RT-101 =====

Route ID : RT-101

Origin : Manila (Cubao)

Destination : Baguio City

Distance : 285.3 km

Duration : 02:00:00

Base Fare : PHP 500.00

===== RT-102 =====

Route ID : RT-102

Origin : Manila (Cubao)

Destination : Batangas Port

Distance : 120.5 km

Duration : 01:30:00

Base Fare : PHP 350.00

===== RT-103 =====

Route ID : RT-103

Origin : Manila (Cubao)

Destination : Naga City

Distance : 350.0 km

Duration : 03:00:00

Base Fare : PHP 600.00

===== RT-104 =====

Route ID : RT-104

Origin : Manila (Cubao)

Destination : La Union (San Juan)

Distance : 400.0 km

Duration : 04:00:00

Base Fare : PHP 700.00

Interactive Seat Selection Matrix

Regular (Real) Priority (GAP)

FRONT OF BUS - DRIVER CAB

01	02	03	04
05	06	07	08
09	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24

0 seats selected. Please click on available.

Passenger & Concessions

Passenger Full Name

Contact Number (e.g. 09123456789)

Passenger Classification
Regular / OSCA (*)

Concession / OSCA (*)

Fare Quotation

Base Fare: PHP 500

Concession Discount

Total Amount: 0

Confirm & Issue Ticket

Clear Form

System Ready. Connected to SQLite Database.

02. Priority Seat & Concession

Seat #01 selected, Senior Citizen OSCA ID verified, 20% concession quoted.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics 4. Passenger & Concessions

1. Select Route & Trip
Transit Route: **BT-301 - Manila (Cubao) to Baguio City (08:00 AM)**
Travel Date (YYYY-MM-DD): **2026-09-26**
Update Seat Availability

Route & Bus Details
Route ID : BT-301
Origin : Manila (Cubao)
Destination : Baguio City
Distance : 245.8 km
Departure : 08:00 AM
Arrival : 09:00 AM
Bus ID : 800-05
Bus Name : SmartBus AC
Capacity : 24 passenger seats
Priority: Senior Citizen (seats: 1, 2, 3, 4)

2. Interactive Seat Selection Matrix
Regular (Red) Priority (Green)
[FRONT OF BUS - DRIVER CAB]
01 02 03 04 05
06 07 08 09 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25
1 Seat #01 (PRIORITY ROW) • Fare: PHP 464.00

3. Passenger & Concessions
Passenger Full Name: **Carson Agapio**
Contact Number (e.g. 0915554421)
Passenger Classification: **Senior Citizen**
Concession / OSCA ID: **OSCA-8819**
Fare Quotation: **Base Fare: PHP 580**
Concession Discount: **20%**
Total Amount: **PHP 464.00**
Confirm & Issue Ticket
Clear Form

System Ready. Connected to SQLite Database.

03. Issued Ticket Confirmation

Atomic SQLite write, seat marked occupied [X], status bar updated.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics 4. Passenger & Concessions

1. Select Route & Trip
Transit Route: **BT-301 - Manila (Cubao) to Baguio City (08:00 AM)**
Travel Date (YYYY-MM-DD): **2026-09-26**
Update Seat Availability

Route & Bus Details
Route ID : BT-301
Origin : Manila (Cubao)
Destination : Baguio City
Distance : 245.8 km
Departure : 08:00 AM
Arrival : 09:00 AM
Bus ID : 800-05
Bus Name : SmartBus AC
Capacity : 24 passenger seats
Priority: Senior Citizen (seats: 1, 2, 3, 4)

2. Interactive Seat Selection Matrix
Regular (Red) Priority (Green)
[FRONT OF BUS - DRIVER CAB]
01 02 03 04 05
06 07 08 09 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25
1 Seat #01 (PRIORITY ROW) • Fare: PHP 464.00

3. Passenger & Concessions
Passenger Full Name: **Carson Agapio**
Contact Number (e.g. 0915554421)
Passenger Classification: **Senior Citizen**
Concession / OSCA ID: **OSCA-8819**
Fare Quotation: **Base Fare: PHP 580**
Concession Discount: **20%**
Total Amount: **PHP 464.00**
Confirm & Issue Ticket
Clear Form

System Ready. Connected to SQLite Database.

Booking Successful! Ticket TKT-1004 issued with Ref: BTK-3A7D4.

04. Tab 2 Manage Bookings

Master ticket list with search filters and status badges.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics 4. Passenger & Concessions

Search Ticket / Ref Code: Find Ticket Reset / View All Filter Status: All Statuses

Active Ticket Dossier & Lifecycle Actions
Select a ticket from the table below to inspect details or execute lifecycle actions.
Cancel Ticket & Refund
View Boarding Pass

All Issued Tickets & Master Transactions

Ticket ID	Reference Code	Passenger Name	Category	Route	Travel Date	Seat	Amount Paid	Status
TKT-1004	BTK-3A7D4	Carson Agapio	Senior Citizen	Manila (Cubao) → Baguio	2026-09-26	#01	PHP 464.00	Confirmed
TKT-1003	BTK-1D7B6	Ona Francis	PWD	Manila (Bundard) → Baguio	2026-09-26	#02	PHP 580.00	Confirmed
TKT-1002	BTK-4C3E2	Juan Dela Cruz	Regular	Manila (Cubao) → Baguio	2026-09-26	#07	PHP 580.00	Confirmed
TKT-1001	BTK-8F3A29	Maria Santos	Senior Citizen	Manila (Cubao) → Baguio	2026-09-26	#01	PHP 464.00	Confirmed

Booking Successful! Ticket TKT-1004 issued with Ref: BTK-3A7D4.

05. Ticket Dossier Inspection

Inspection card with SHA-256 verification and priority seat validation.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics 4. Passenger & Concessions

Search Ticket / Ref Code: Find Ticket Reset / View All Filter Status: All Statuses

Active Ticket Dossier & Lifecycle Actions
Ticket ID : TKT-1004
Ticket-Proof Ref : BTK-8F3A29 (SHA-256 HMAC VERIFIED)
Booking Status : CONFIRMED
Passenger Name : Maria Santos
Classification : Senior Citizen
Concession ID Proof : OSCA-7785
Contact Number : 0915554421
Cancel Ticket & Refund
View Boarding Pass

All Issued Tickets & Master Transactions

Ticket ID	Reference Code	Passenger Name	Category	Route	Travel Date	Seat	Amount Paid	Status
TKT-1004	BTK-3A7D4	Carson Agapio	Senior Citizen	Manila (Cubao) → Baguio	2026-09-26	#01	PHP 464.00	Confirmed
TKT-1003	BTK-1D7B6	Ona Francis	PWD	Manila (Bundard) → Baguio	2026-09-26	#02	PHP 580.00	Confirmed
TKT-1002	BTK-4C3E2	Juan Dela Cruz	Regular	Manila (Cubao) → Baguio	2026-09-26	#07	PHP 580.00	Confirmed
TKT-1001	BTK-8F3A29	Maria Santos	Senior Citizen	Manila (Cubao) → Baguio	2026-09-26	#01	PHP 464.00	Confirmed

Booking Successful! Ticket TKT-1004 issued with Ref: BTK-3A7D4.

06. Official Boarding Pass

High-fidelity modal dialog for terminal commuter boarding.

SMARTBUS TRANSIT — OFFICIAL BOARDING PASS

GER NAME : MARIA SANTOS
RY : SENIOR CITIZEN

URE ROUTE: MANILA (CUBAO) -> BAGUIO CITY
DATE : 2026-09-26 | TIME: 08:00 AM
/ BUS : DLTB-8012 (Standard AC)
ED SEAT : SEAT #01 [PRIORITY]

TY CODE : BTK-8F3A29 (SHA-256 Verified)
STATUS : CONFIRMED
AID : PHP 464.00

arrive at terminal 30 minutes before scheduled depa

Test ID	Test Method Name	Category	Verification Criteria	Status
TC-01	test_01_singleton_database_instance	Architecture	Verifies BusTicketingDatabase enforces Singleton pattern.	PASS
TC-02	test_02_sqlite_persistence_round_trip	Persistence	Verifies SQLite CRUD operations for Routes and Buses.	PASS
TC-03	test_03_route_validation_boundaries	NFR-2	Verifies positive distance and non-negative base fare constraints.	PASS
TC-04	test_04_bus_seating_and_priority_validation	NFR-2	Verifies seating bounds and priority seat designations.	PASS
TC-05	test_05_passenger_contact_format_validation	NFR-1	Verifies mobile contact regex sanitization.	PASS
TC-06	test_06_mandatory_concession_id_validation_nfr1	NFR-1	Verifies mandatory concession ID check for Senior, PWD, and Student.	PASS
TC-07	test_07_senior_priority_seat_and_concession_discount	Unique UC	Verifies Row 1 priority seat booking with statutory 20% discount.	PASS
TC-08	test_08_regular_passenger_full_fare	FR-1	Verifies regular passengers receive standard undiscounted fare.	PASS
TC-09	test_09_tamper_resistant_reference_code_format_nfr1	NFR-1	Verifies cryptographically salted SHA-256 reference codes.	PASS

TC-10	test_10_atomic_double_booking_prevention_nfr2	NFR-2	Verifies duplicate seat booking raises ValueError & rolls back.	PASS
TC-11	test_11_ticket_cancellation_releases_seat_fr2	FR-2	Verifies cancelled ticket releases seat for immediate re-booking.	PASS
TC-12	test_12_analytics_calculation	Metrics	Verifies gross revenue, ticket counters, and subsidy calculations.	PASS

9. Installation & Execution Guide

Prerequisites

- Python 3.10+ (Standard distribution)
- SQLite3 (Included natively in Python standard library)
- Tkinter & Pillow (**pip install pillow**)

Running the Application

Launch the graphical interface pre-loaded with sample routes, buses, and bookings:

Code Block: POWERSHELL

```
python main.py
```

Running the Unit Test Suite

Execute the automated test suite with verbose output:

Code Block: POWERSHELL

```
python -m unittest test_system.py -v
```

Capturing Automated High-Resolution Screenshots

Run the headless capture suite to regenerate all 9 screenshots into **screenshots/**:

Code Block: POWERSHELL

```
python capture_screenshots.py
```

10. Project Directory Tree

Code Listing

```
Practical/
■■■ models/
■ ■■■ __init__.py # Re-exports all domain entities and database singleton
■ ■■■ bus.py # Bus entity with cabin configuration & priority seats
■ ■■■ database.py # SQLite BusTicketingDatabase singleton with ACID queries
■ ■■■ factory.py # BusFactory & TicketFactory with concession calculations
■ ■■■ passenger.py # Passenger entity with PII & concession ID validation
■ ■■■ route.py # Route entity with travel metrics and pricing
■ ■■■ ticket.py # Ticket entity, TicketStatus enum, & SHA-256 ref generator
■■■ screenshots/ # 9 Programmatic high-resolution PNG screen captures
■ ■■■ 01_tab_booking_seat_map.png
■ ■■■ 02_selected_priority_seat.png
■ ■■■ 03_issued_ticket.png
■ ■■■ 04_tab_manage_bookings.png
■ ■■■ 05_ticket_dossier_inspection.png
■ ■■■ 06_boarding_pass_modal.png
■ ■■■ 07_ticket_cancellation_seat_released.png
■ ■■■ 08_tab_system_analytics.png
■ ■■■ 09_unit_tests_pass.png
■■■ app.py # Main Tkinter GUI application (3 tabs, visual seat map)
■■■ bus_ticketing.db # SQLite persistent database file
■■■ capture_screenshots.py # Automated screenshot generation suite
■■■ main.py # System entry point with sample data pre-population
■■■ test_system.py # Complete 12-test unittest suite
■■■ README.md # Complete project documentation and specifications
```